

SELECT Queries

Learning Objectives

- Write SELECT statements to retrieve data from a single table
- Use SELECT DISTINCT to eliminate duplicate records
- Apply column aliases and ORDER BY to format query results

Engage (5-7 minutes)

A teacher wants to see only the names and grades of students in Class 10, sorted alphabetically. This is exactly what a SELECT query does-it retrieves only the columns you need from the rows that match your criteria.

Explore (10-12 minutes)

Given a “Students” table with 20 records, write SQL queries to: select all columns, select only FirstName and Grade, select only distinct Grade values, and sort results by LastName in descending order. Test each query in the practice environment.

Explain (10-12 minutes)

The SELECT statement retrieves data from a table. Use SELECT * to get all columns, or list specific columns separated by commas. SELECT DISTINCT removes duplicate values. AS creates column aliases for readability. ORDER BY sorts results - ASC for ascending (default) and DESC for descending.

Elaborate (8-10 minutes)

A school admin wants a report showing student names and grades, sorted by grade then by name within each grade. Write the SQL query using ORDER BY with multiple columns. Explain how the database processes this sorting.

Evaluate (5-8 minutes)

Write a SELECT query to retrieve City and Population from a “Cities” table, showing only distinct cities sorted by Population descending. Explain the difference between SELECT and SELECT DISTINCT.